MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. MAKE-UP EXAMINATIONS DECEMBER 2019

SUBJECT: BIOREMEDIATION [BIO 4001]

Date of Exam: **/12/2019** Time of Exam: Max. Marks: **50**

Instructions to Candidates:

- Answer ALL the questions & missing data may be suitable assumed

1A.	How are bioremediation methods advantageous than conventional methods of waste treatment and disposal?	**2m**
1B.	What are the stages in coal gasification? What is the application of the final product, obtained in the process?	**4m**
1C.	What do mean by porosity of soil? How is it measured mathematically? What is its usual range for normal soils?	**4m**
2A.	How is the free energy change used to correlate the energy released in a reaction?	**2m**
2B.	Explain about the acclimation phase in the life cycle of a bacterium growing in the presence of a polycyclic aromatic hydrocarbon.	**3m**
2C.	How will moisture and microbial factors influence biodegradation of a toxic compound?	**5m**
3A.	What methods are used to alter the pH of soil? Explain with examples for all the methods.	**3m**
3B.	What do you understand by the electron tower concept? How is it used for bioremediation?	**3m**
3C.	Explain the mechanism by which aromatic compounds are enzymatically degraded. Draw the relevant pathway for a representative compound.	**4m**
4A.	How are soil samples taken out from a site during treatment using a Land Treatment Unit? Explain using a diagram.	**4m**
4B.	Name any two design parameters for a Pump, Treat and Reinjection system. Explain how they influence the degradation process.	**6m**
5A.	Enlist two advantages and disadvantages of phytoremediation.	**4m**
5B.	Consider the metabolism of glucose by aerobic microbes, for the redox pairs CO_2/glucose and O_2/H_2O. Write the half reactions and the combined balanced reactions. Find the free energy for this reaction if the RP values for O_2/H_2O is +0.82V and that for $CO_2/C_6H_{12}O_6$ is –0.43V.	**6m**